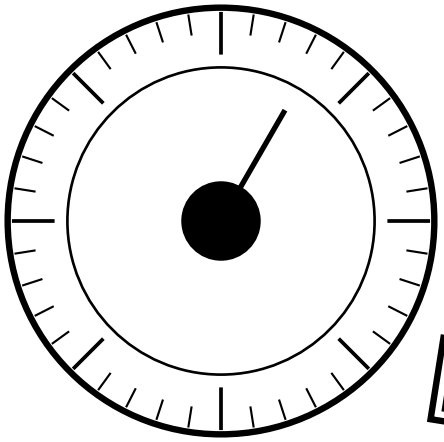
BONUS MISSION PACK

Ten extra missions for agents who finished the book,
or who just want more. Missions 81 to 90.
Hints, check codes and full solutions are inside.



FOR AGENTS ONLY

FUN BOOKS
PUBLISHER

LOGIC GRID | LEVEL ■■□ ADVANCED TYPICAL TIME: 16 MIN

THE HARBOR SIGN-OUT

The Director wants a clean roster before the lab inspection this afternoon. Match each of the four agents with every one of their details.

RULES | Each agent matches exactly one item in every category, and no two agents share an item. Earlier, later, lower and higher mean by any amount unless the clue says exactly. A clue "If A, then B" means that A never happens without B.

AGENT	ROBOT				COURSE				SHIFT START			
	Cog	Sprocket	Gear	Pixel	Ciphers	Botany	Rockets	Optics	7 AM	8 AM	9 AM	10 AM
Elena												
Nadia												
Zoe												
Malik												

SHIFT START	
7 AM	
8 AM	
9 AM	
10 AM	

COURSE	
Ciphers	
Botany	
Rockets	
Optics	

1. The agent training in Ciphers starts later than the agent with the robot Sprocket.
2. The agent who starts at 9 AM is training in either Ciphers or Optics.
3. Nadia starts earlier than the agent training in Optics.
4. The agent who starts at 8 AM does not work with the robot Cog.
5. Nadia is training in Botany.
6. Zoe works with the robot Pixel.
7. The agent training in Optics works with either the robot Pixel or the robot Gear.
8. The agent who starts at 9 AM works with the robot Pixel.
9. The agent training in Optics does not start at 9 AM.
10. Elena does not work with the robot Cog.
11. Malik is not training in Rockets.

Which agent works with the robot Sprocket?

ANSWER _____ CHECK CODE (PAGE 12) _____

SHIFT CIPHER | LEVEL ■■■ ELITE TYPICAL TIME: 21 MIN

CODE NAME: QUIET COMET

Theo found this card taped under a desk in the Map Room. Crack it before anyone else does. The shift is the sum of the digits in the badge number 773.

RULES | Every letter was moved forward by the same secret number of places. Use the riddle in the brief to find that number, then move every letter back by it. Before A, go around to Z.

INTERCEPTED MESSAGE

Q	F	V		W	Z	O	V	U		K	Y	V

K	Z	E	P		T	F	D	G	R	J	J		R	K

K	V	E

DECODER KEY

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

After Z comes A again. Before A comes Z.

WORK SPACE

What is the fourth word of the message?

ANSWER _____ CHECK CODE (PAGE 12) _____

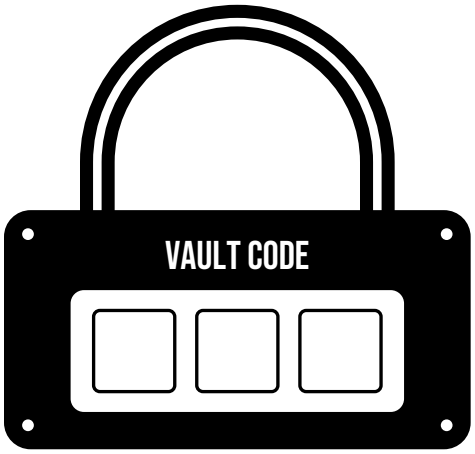
LOCK CODE | LEVEL ■■□ ADVANCED TYPICAL TIME: 10 MIN

THE RUSTY LAB SAFE

A rival lab tried to crack the mirror map vault and left four failed guesses behind. Use them to open it first.

RULES | The code never repeats a digit, and every clue below is true. Each clue counts the guess digits in the code, and how many sit in the right spot. Any other digit of that guess is not in the code.

	GUESS	WHAT THE LOCK SAYS
1	0 9 3	One digit is correct but in the wrong spot.
2	7 4 6	One digit is correct but in the wrong spot.
3	7 3 5	Two digits are correct: one in the right spot, one in the wrong spot.
4	8 2 5	Nothing is correct.



SPOT TRACKER

Cross out a digit at the top once it is out.
Put an X where a digit cannot go.

DIGIT	0	1	2	3	4	5	6	7	8	9
1ST										
2ND										
3RD										

What is the 3-digit code?

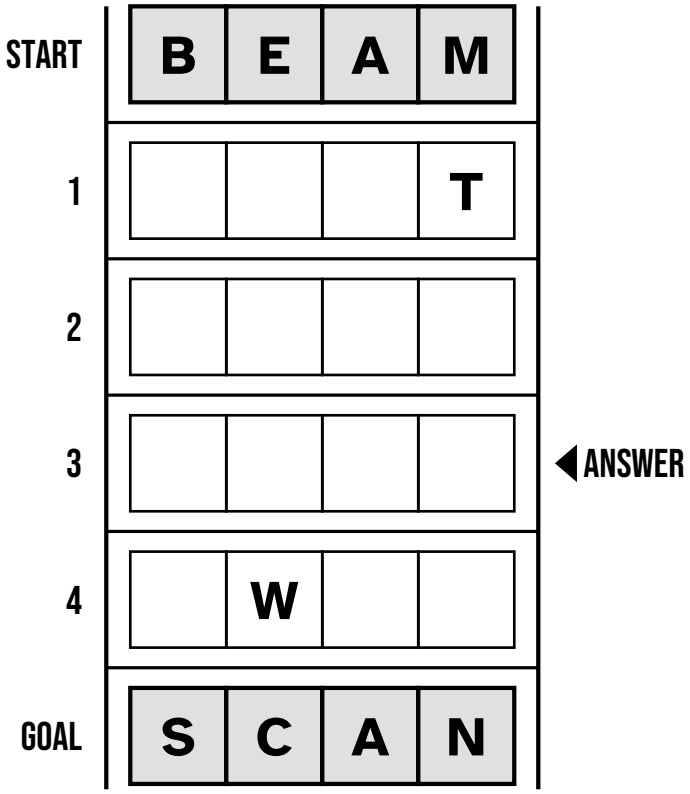
ANSWER _____ CHECK CODE (PAGE 12) _____

WORD LADDER | LEVEL ■■□ ADVANCED TYPICAL TIME: 9 MIN

SIGNAL PATH: BEAM TO SCAN

A message drone dropped half its cargo. Use the printed letters to rebuild the climb from BEAM to SCAN.

RULES | Change exactly one letter at each step to climb from the top word to the bottom word. Every rung must be a real English word, and no word can appear twice. Printed letters are already in place.



WORK SPACE

What word sits on rung 3 of the ladder?

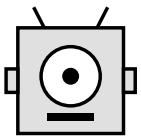
ANSWER _____ CHECK CODE (PAGE 12) _____

TRUTH OR LIE | LEVEL ■■□ ADVANCED TYPICAL TIME: 10 MIN

THE MARBLE VAULT

The lab robots at this checkpoint were reprogrammed, and some of them now lie. Work out which ones you can trust before you go any further.

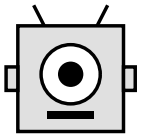
RULES Each character is a truth-teller, whose statement is true, or a liar, whose statement is false. Same type means both truth-tellers or both liars, and an "and" statement needs both parts true. "Us" means everyone on this page, the speaker too; "at least one" means one or both.



Diego and I are the same type.

T ☐ L ☐

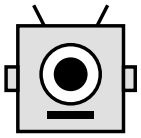
JAMAL



Jamal and I are different types.

T ☐ L ☐

HAZEL



Jamal and Hazel are different types.

T ☐ L ☐

DIEGO

WORK SPACE

Who is the only liar?

ANSWER ☐ CHECK CODE (PAGE 12) ☐

NONOGRAM | LEVEL ■■□ ADVANCED TYPICAL TIME: 10 MIN

THE SHADOW PRINTOUT

Agent Amara scanned a sealed crate, but the printout came back as numbers. Rebuild the scan.

RULES Each number is a run of filled squares in that row or column, in the order shown. Runs are separated by at least one empty square. A 0 means the whole line stays empty. Fill the right squares to reveal a hidden picture.

	2	4	6	12	14	92	92	14	12	6	4	2
4												
2 2 2												
2 2 2												
3 2 3												
3 2 3												
2 2 2												
2 2 2												
1 2 1												
4												
1 1												
4												
4												

What does the picture show? Write it in one word (7 letters).

ANSWER _____ CHECK CODE (PAGE 12) _____

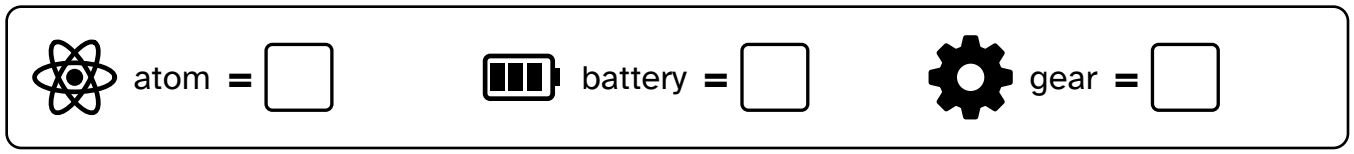
SYMBOL BALANCE | LEVEL ■■□ ADVANCED TYPICAL TIME: 9 MIN

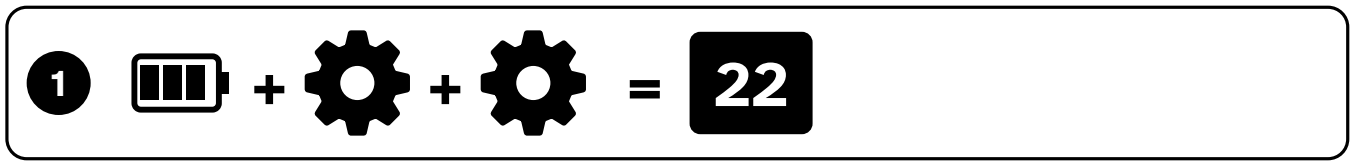
THE SPARK CHECK

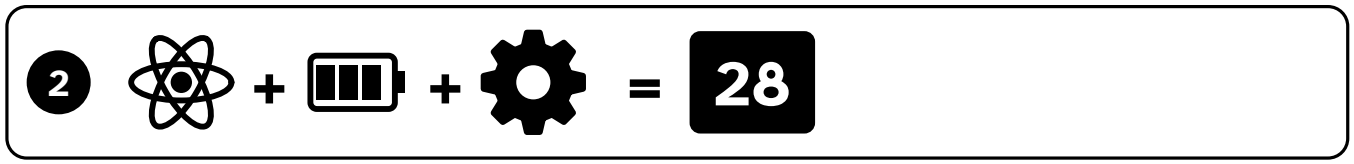
A shipping crate for the heat glove arrived with symbol labels. Sofia needs the final count before the meeting.

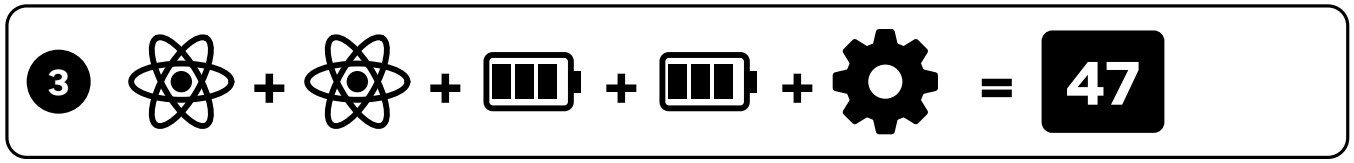
RULES | Each symbol stands for a whole number from 1 to 20. The same symbol always has the same value on every line. Every line balances: the left side equals the number on the right.

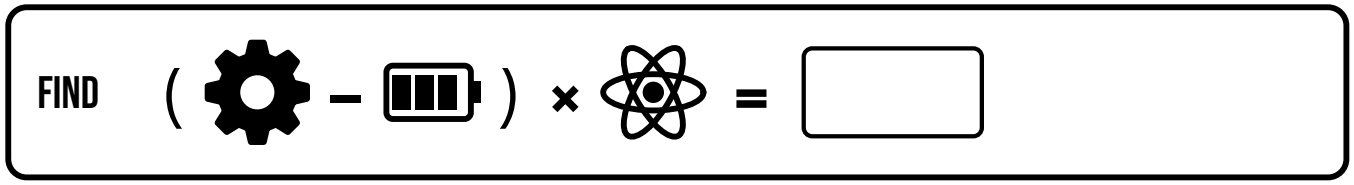
SYMBOL KEY











WORK SPACE

What is (gear – battery) × atom?

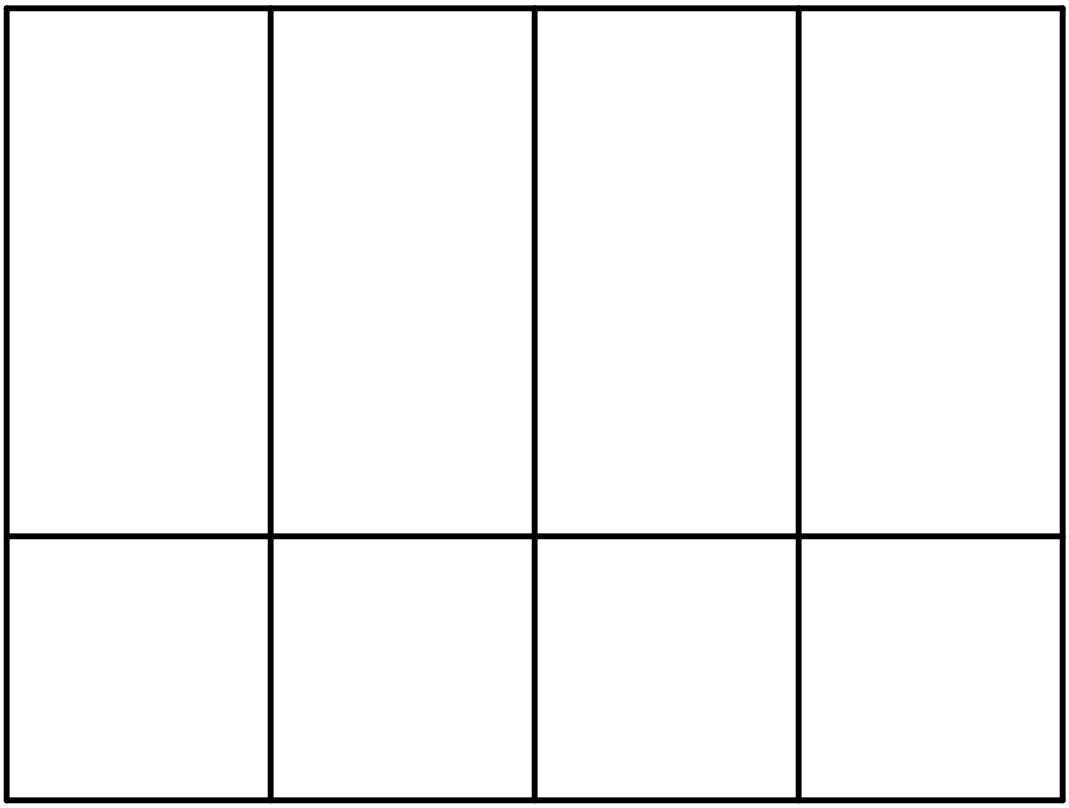
ANSWER □ CHECK CODE (PAGE 12) □

SHAPE COUNT | LEVEL ■■□ ADVANCED TYPICAL TIME: 11 MIN

CROSSED LATTICE

A security grid protects the quiet bike motor. Count every rectangle in it, big and small.

RULES | Count every shape of the kind asked: small or large, upright or tilted. Shapes can overlap and can be made of smaller pieces. A square also counts as a rectangle.



TALLY BY SIZE ☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

Size = how many small pieces a shape covers. Use only the boxes you need.

How many rectangles are in the figure, squares included? Write the number.

ANSWER ☐ **CHECK CODE (PAGE 12)** ☐

WORD MATH | LEVEL ■■□ ADVANCED TYPICAL TIME: 12 MIN

DIGITS IN PROOF

The robot only accepts numbers, but the memo is in letters. Translate the sum for it.

RULES | Each letter stands for one digit from 0 to 9, and different letters stand for different digits. No number starts with 0. Make the addition true.

	G	A	U	G	E
+			K	E	Y
P	R	O	O	F	

CODE KEY

A	E	F	G	K	O	P	R	U	Y
								2	

DIGITS USED (CROSS THEM OUT)

0	1	2	3	4	5	6	7	8	9

WORK SPACE

What number does PROOF stand for?

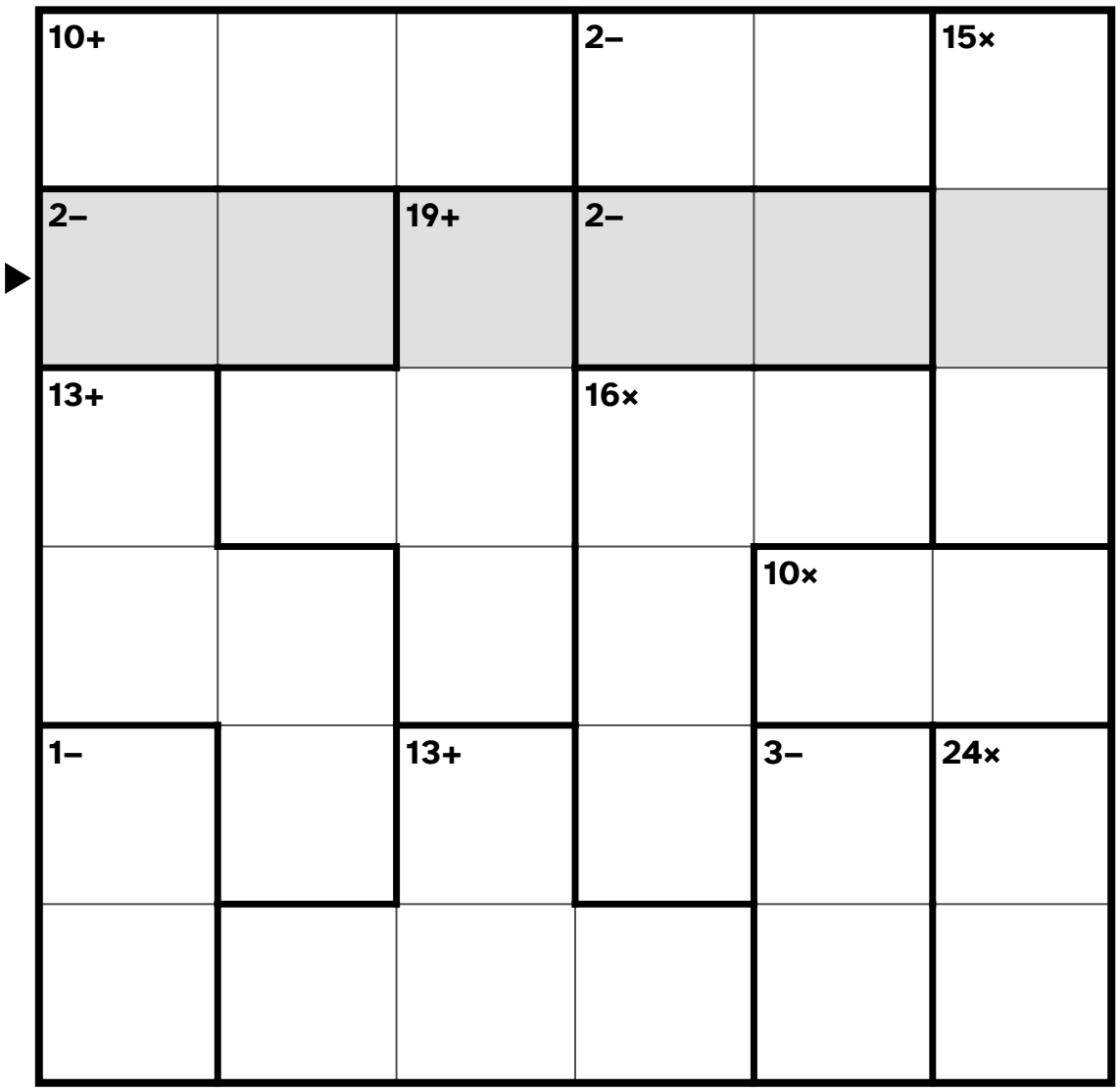
ANSWER _____ CHECK CODE (PAGE 12) _____

CALC CAGES | LEVEL ■■■ ELITE TYPICAL TIME: 20 MIN

THE SILVER GATE

Someone swapped the numbers on the greenhouse panel. Put them back in their cages.

RULES Each row and each column uses every number from 1 to the grid size exactly once. A thick outline is a cage: its numbers must make the target with the sign shown. For a minus or divide cage, start with the bigger number. A number can repeat inside a cage if the two are not in the same row or column.



Read the shaded row from left to right. What number do you get?

ANSWER CHECK CODE (PAGE 12)

CHECK YOUR ANSWER

Turn your answer into a check code. Then look for your code in the list below.

If your code is in the list, your answer is right. Nobody had to show you the solution.

1. Write each letter as its number: A = 1, B = 2, C = 3 … Z = 26. A digit stays as it is.
2. Multiply the first by 1, the second by 2, the third by 3, and so on.
3. Add up the results. Then add ten times the mission number.

EXAMPLE · ANSWER "GEAR" IN MISSION 4

G	E	A	R
7 x 1 = 7	5 x 2 = 10	1 x 3 = 3	18 x 4 = 72

7 + 10 + 3 + 72 = 92

92 + 40 (mission 4 x 10) = 132

Check code: 132

CODES FOR MISSIONS 81 TO 90

Half of these codes are decoys.

824	**858**	**883**	**887**	**915**
917	**921**	**965**	**967**	**988**
995	**1000**	**1020**	**1080**	**1149**
1170	**1197**	**1211**	**1243**	**1259**

Code not in the list? Check your arithmetic first. If the code is still missing, go back to the mission: one step went wrong somewhere. Take a hint if you want one.

HINT A

Read only the hint for your mission. Cover the rest with a sheet of paper.

DEPT 9 · BONUS FILES

M81 Mark every direct clue first. Then look hard at clue 2: it is the key, and the Either-Or technique unlocks it.

M82 Solve the riddle in the brief first. Its answer is a number of places.

M83 Start with the clue that says nothing is correct, and cross out its digits everywhere.

M84 Start at the top: change one letter of BEAM so it matches the printed letters on rung 1.

M85 Start with the statement made by Jamal. Try the Two-Way Check technique on it.

M86 Start with the biggest clues, such as column 6 and column 7. Long runs in a short line leave little room to slide.

M87 Find two lines where one holds everything the other has, plus a little more.

M88 Sort the rectangles by how many small squares wide and tall they are.

M89 Look at the thousands column first and list the digits that could work there.

M90 Look for the cage with the fewest ways to make its target. Small cages are a good start.

HINT B

Try Hint A first. Hint B does the first step for you, never the whole mission.

DEPT 9 · BONUS FILES

M81 **[Direct Clue]** Clue 5 gives a match: Nadia is training in Botany. Mark the O and X the rest of its row and column.

M82 The riddle gives a shift of 17, so the first secret letter, Q, becomes Z.

M83 Clue 4 (825): 2, 5 and 8 are out.

M84 First step done: rung 1 is BEAT.

M85 **[Two-Way Check]** Test Jamal both ways: as a truth-teller or as a liar, the result is that Diego is a truth-teller.

M86 Row 4 (clue 3 2 3) is empty. Slide the runs both ways, then fill squares 3 and 10.

M87 Line 3 holds everything in line 2 plus an atom and a battery, so atom + battery = 47 – 28 = 19.

M88 There are 4 rectangles that are 1 by 1.

M89 First step done: A = 9.

M90 The 24× cage at row 5, column 6 can only use {4, 6}.

Each solution names the technique it uses in bold. Learn the names: you will meet the same techniques again in harder missions.

MISSION 81 THE HARBOR SIGN-OUT

ANSWER: Elena

Agent	Robot	Course	Shift Start
Elena	Sprocket	Rockets	8 AM
Nadia	Cog	Botany	7 AM
Zoe	Pixel	Ciphers	9 AM
Malik	Gear	Optics	10 AM

1. **[Direct Clue]** Clue 4 rules out a box: the agent with the robot Cog does not start at 8 AM. Mark every other clue that names a match or a non-match the same way.
2. **[Either-Or]** Clue 2 leaves only two courses for the agent who starts at 9 AM, so every other box in that row gets an X.
3. **[Last Box]** Only one box is left in that row of robots, so Elena works with the robot Sprocket.
4. **[Order Bounds]** Clue 1 sets an order, so the agent with the robot Sprocket does not start at 10 AM.
5. Finish the grid with the same moves. The answer is Elena.

MISSION 82 CODE NAME: QUIET COMET

ANSWER: TINY

Q F V W Z O V U K Y V
Z O E F I X E D T H E
K Z E P T F D G R J J
T I N Y C O M P A S S
R K K V E
A T T E N

Key: shift back 17. Answer: TINY.

1. **[Crack the Key]** 7 + 7 + 3 = 17, so the shift is 17.
2. **[Shift Back]** Move every letter back 17 places: Q becomes Z, and so on.
3. **[Word Check]** The message reads: ZOE FIXED THE TINY COMPASS AT TEN. Every word is a real word, so the fourth word is TINY.

MISSION 83 THE RUSTY LAB SAFE

ANSWER: 137

1 [0][9][3] 1 wrong spot

2 [7][4][6] 1 wrong spot

3 [7][3][5] 1 right spot, 1 wrong spot

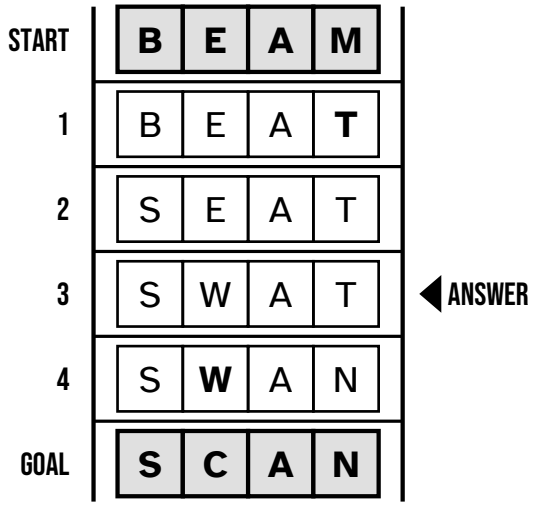
4 [8][2][5] nothing correct

CODE [1][3][7]

1. **[Nothing Is Correct]** Clue 4 (825): 2, 5 and 8 are out. **[Count the Digits]** Clue 3 (735): 3 and 7 are in the code.
2. **[Count the Digits]** Clue 1 (093): 0 and 9 are out. **[Count the Digits]** Clue 2 (746): 4 and 6 are out.
3. **[Wrong Spot]** Clue 1 (093): 3 is not in the third spot. **[Wrong Spot]** Clue 2 (746): 7 is not in the first spot.
4. **[Place Count]** Clue 3 (735): 3 goes in the second spot. **[Only Spot Left]** 7 goes in the third spot.
5. **[Only Spot Left]** Only 1 can still fill the first spot.
6. So the code is 137.

MISSION 84 SIGNAL PATH: BEAM TO SCAN

ANSWER: SWAT



1. **[Step Down]** Change one letter of BEAM: only BEAT fits the printed letters on rung 1.
2. **[Step Up]** Work up from SCAN: only SWAN fits the printed letters on rung 4.
3. **[Count the Gap]** Next to SWAN, every other choice is too far from BEAT, so rung 3 is SWAT.
4. **[Squeeze Play]** Rung 2 must be one letter from BEAT and from SWAT, so it is SEAT.
5. The word on rung 3 is SWAT.

MISSION 85 THE MARBLE VAULT

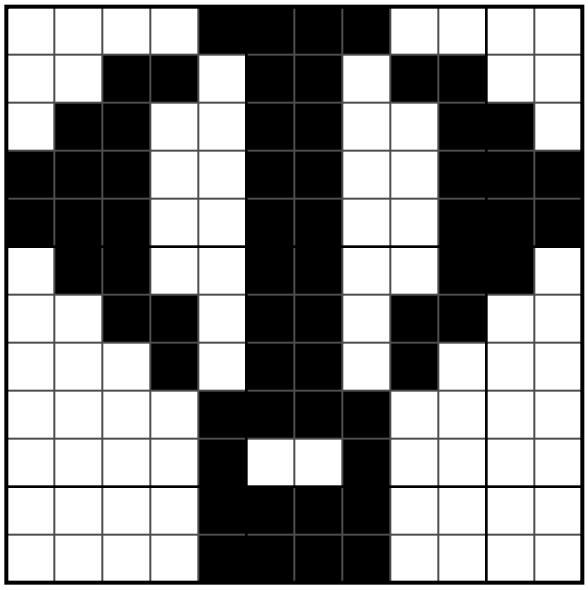
ANSWER: Jamal

Jamal liar
Hazel truth-teller
Diego truth-teller

1. **[Two-Way Check]** Test Jamal both ways: as a truth-teller or as a liar, the result is that Diego is a truth-teller.
2. **[Two-Way Check]** Test Hazel both ways: as a truth-teller or as a liar, the result is that Jamal is a liar.
3. **[Pair Logic]** Diego is a truth-teller, so the statement about the pair is true. That means Hazel is a truth-teller.
4. The answer is Jamal.

MISSION 86 THE SHADOW PRINTOUT

ANSWER: BALLOON



1. **[Overlap]** Row 4 (clue 3 2 3) is empty. Slide the runs both ways, then fill squares 3 and 10.
2. **[Match the Runs]** In row 2 (clue 2 2 2), match each marked square to its run. Then mark squares 5 and 8 empty.
3. **[Line Done]** Column 3 (clue 6) has all its filled squares, so mark square 8 empty.
4. Keep switching between rows and columns. Each new square unlocks another line.
5. The finished picture shows a balloon.

MISSION 87 THE SPARK CHECK

ANSWER: 75

atom = 15 battery = 4 gear = 9

1 + + = **22**

2 + + = **28**

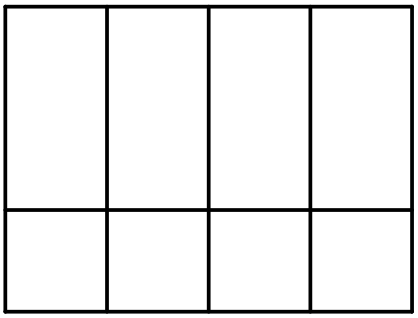
3 + + + + = **47**

? (–) × = **75**

1. **[Compare Scales]** Line 3 holds everything in line 2 plus an atom and a battery, so atom + battery = 47 – 28 = 19.
2. **[Compare Scales]** Line 2 holds everything in atom + battery = 19 plus a gear, so gear = 28 – 19 = 9.
3. **[Swap In]** Put gear = 9 into line 1: battery + 9 + 9 = 22, so the battery is 4.
4. **[Swap In]** Put battery = 4 and gear = 9 into line 2: atom + 4 + 9 = 28, so the atom is 15.
5. So (gear – battery) × atom = (9 – 4) × 15 = 75.

MISSION 88 CROSSED LATTICE

ANSWER: 30



Answer: 30 rectangles. Sort the rectangles by size, counting in small squares: 1 by 1: 4; 1 by 2: 7; 1 by 3: 6; 1 by 4: 1; 2 by 2: 3. The bigger ones: 2 by 3: 5; 2 by 4: 1; 3 by 3: 2; 3 by 4: 1.

1. **[Width x Height]** Sort the rectangles by size, counting in small squares: 1 by 1: 4; 1 by 2: 7; 1 by 3: 6; 1 by 4: 1; 2 by 2: 3.
2. **[Width x Height]** The bigger ones: 2 by 3: 5; 2 by 4: 1; 3 by 3: 2; 3 by 4: 1.
3. Add them all: 4 + 7 + 6 + 1 + 3 + 5 + 1 + 2 + 1 = 30 rectangles.

MISSION 89 DIGITS IN PROOF

ANSWER: 50113

	G 4	A 9	U 2	G 4	E 6
+			K 8	E 6	Y 7
	P 5	R 0	O 1	O 1	F 3

A=9, E=6, F=3, G=4, K=8, O=1, P=5, R=0, U=2*, Y=7

* given

1. Start with the given letters: U = 2.
2. **[Carry Chain]** Checking the ones and tens columns together leaves E = 6.
3. **[Carry Chain]** Checking the ones and tens columns together leaves F = 3.
4. **[Carry Chain]** Checking the ones and tens columns together leaves G = 4.
5. Keep using the same checks for the rest: A = 9, R = 0, O = 1, K = 8, Y = 7, P = 5.
6. So PROOF = 50113.

MISSION 90 THE SILVER GATE

ANSWER: 642315

10+ 5	2	3	2– 6	4	15× 1
2– 6	4	19+ 2	2– 3	1	5
13+ 4	6	5	16× 1	2	3
3	1	6	4	10× 5	2
1– 1	5	13+ 4	2	3– 3	24× 6
2	3	1	5	6	4

1. **[Cage Options]** The 24× cage at row 5, column 6 can only use {4, 6}.
2. **[Cage Claim]** Every option for the 16× cage at row 3, column 4 puts a 1 in column 4, so no other square there holds a 1.
3. **[Line Total]** Row 1 must add up to its full total, so the open squares there must add up to 11.
4. **[Test and Reject]** For the 13+ cage at row 5, column 3, the set {1, 3, 4, 5} leaves a number with nowhere to go, so cross it out.
5. Keep going with Last Option and Only Spot until every square is full.
6. The shaded row reads 642315.